

Beyond Quadratics

1)

Solve the inequality $(2x - 1)(x + 1)(x + 7) > 0$ by considering what values of x make each factor in the product positive.

2)

In this problem, we solve the inequality $\frac{(x + 3)(1 - x)}{x - 5} \geq 0$. Remember: division by 0 is illegal!

3)

In this problem, we solve the inequality $\frac{2}{x - 2} - \frac{2}{x + 3} > -1$.

- (a) Move all terms to one side and write that side with a common denominator.
- (b) For what values of x is the numerator positive? For what values of x is the denominator positive?
- (c) Solve the inequality.

4)

Solve the inequality $(2x - 3)(x^2 + 8x + 15) \leq 0$.

5)

What values of r satisfy the inequality $(r - 3)^2(4r^2 - 25) > 0$?

6)

Solve the inequality $\frac{1}{x - 7} - \frac{2x}{1 - x} \geq 2$.

CHALLENGE

Solve the inequality

$$\frac{(x + 3) + \frac{1}{x+3}}{(x - 3)(x + 2)} \geq \frac{2}{(x - 3)(x + 2)}.$$

Answers:

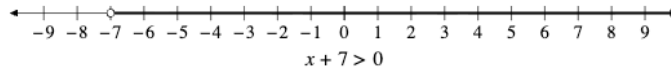
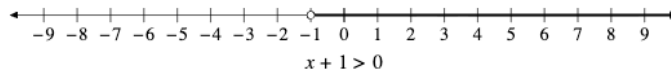
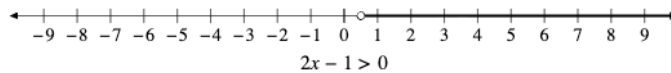
1)

Solution for Problem 15.5: We have three factors whose product must be positive. Therefore, either all three factors are positive, or one is positive and the other two are negative. The factor $2x - 1$ is positive for $x > \frac{1}{2}$, the factor $x + 1$ is positive for $x > -1$, and $x + 7$ is positive for $x > -7$. So, all three are positive when $x > \frac{1}{2}$, and $x + 7$ is positive with the other two negative when $-7 < x < -1$. We can use a table to keep track of the signs of the factors, and of the whole product $(2x - 1)(x + 1)(x + 7)$, for different ranges of values of x :

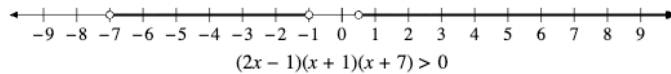
	$2x - 1$	$x + 1$	$x + 7$	$(2x - 1)(x + 1)(x + 7)$
$x > \frac{1}{2}$	+	+	+	+
$-1 < x < \frac{1}{2}$	-	+	+	-
$-7 < x < -1$	-	-	+	+
$x < -7$	-	-	-	-

In the table above, we have a column for each factor in $(2x - 1)(x + 1)(x + 7)$, and a column for the product itself. The factors' columns make determining the sign of the product very easy. From our table, we can quickly see that the product is positive when $x > \frac{1}{2}$ and when $-7 < x < -1$.

We can also find our solutions by considering number lines for each factor, bolded where the factor is positive:



The product of the three factors is positive when all three are positive, or when one is positive and the other two are negative. Using these facts, we can combine the three number lines above to plot the solutions to $(2x - 1)(x + 1)(x + 7) > 0$ on the number line below:



So, our solutions are all values of x such that $x > \frac{1}{2}$ or $-7 < x < -1$. In interval notation, this is $x \in (-7, -1) \cup \left(\frac{1}{2}, \infty\right)$. $\square$

2)

Solution for Problem 15.6: This inequality differs from previous inequalities we've tackled in a couple ways. First, we have a factor in a denominator. Second, we have a factor in which the coefficient of x is negative (the $1 - x$ in the numerator). We can take care of the latter difficulty by multiplying both sides of the inequality by -1 . The right side is still 0. On the left, we have

$$-\frac{(x+3)(1-x)}{x-5} = \frac{(x+3)[-1(1-x)]}{x-5} = \frac{(x+3)(x-1)}{x-5}.$$

Because we are multiplying by a negative number, we must reverse the sign of our inequality, and our inequality becomes

$$\frac{(x+3)(x-1)}{x-5} \leq 0.$$

Now we turn to the issue of having a factor in the denominator. Whether a factor is in the numerator or denominator of an expression does not affect the sign of that expression. Therefore, we can compare

$$\frac{(x+3)(x-1)}{x-5}$$

to zero by considering the signs of $x+3$, $x-1$, and $x-5$. We also have to remember that we cannot have $x=5$, since this will make the denominator zero. However, $x=-3$ and $x=1$ are both solutions to the given nonstrict inequality.

In order for the expression to be negative, we must have all three of $x+3$, $x-1$, and $x-5$ be negative, or have one of them negative and the other two positive. Since $x-5$ is negative for $x < 5$, $x-1$ is negative for $x < 1$, and $x+3$ is negative for $x < -3$, we see that our expression is negative for $x < -3$ (all three negative) and when $1 < x < 5$ ($x-5$ is negative and the others are positive). We could also have organized the information about the signs of $x-5$, $x-1$, and $x+3$ in a table:

	$x-5$	$x-1$	$x+3$	$\frac{(x+3)(x-1)}{x-5}$
$x > 5$	+	+	+	+
$1 < x < 5$	-	+	+	-
$-3 < x < 1$	-	-	+	+
$x < -3$	-	-	-	-

For example, in the $x > 5$ row above, we have recorded that all three of $x-5$, $x-1$, and $x+3$ are positive when $x > 5$. Looking at this table, we can even more clearly see that our expression is negative when $x < -3$ and when $1 < x < 5$.

Recalling that $x = -3$ and $x = 1$ are solutions as well, because the inequality is nonstrict, our solutions are all x such that $x \leq -3$ or $1 \leq x < 5$. In interval notation, this is $x \in (-\infty, -3] \cup [1, 5)$.

WARNING!!



When comparing an expression with factors in the numerator and denominator to zero, be careful not to include any values in your solution set that make the denominator equal to zero.

We could also have solved this problem without doing the first step of multiplying by -1 . Then, we would have to be careful to note that $1-x$ is *negative* for $x > 1$ and *positive* for $x < 1$. See if you can complete this approach on your own. (You better get the same answer we just found!) □

Sometimes we have to do a little work to determine what factors we must consider.

3)

Solution for Problem 15.7: What's wrong with starting this problem like this:

Bogus Solution: We start by multiplying both sides by $(x - 2)(x + 3)$ to get rid of the fractions. This gives us:



$$2(x + 3) - 2(x - 2) > -1(x - 2)(x + 3).$$

The problem with this start is that if $(x - 2)(x + 3)$ is negative, then we must change the direction of the inequality sign.

Fortunately, we know how to compare products and quotients of expressions to zero, so we move all the terms to the left side, leaving 0 on the right:

$$\frac{2}{x - 2} - \frac{2}{x + 3} + 1 > 0.$$

We write the left side with a common denominator so that we'll have a quotient to compare to zero:

$$\frac{2(x + 3)}{(x - 2)(x + 3)} - \frac{2(x - 2)}{(x - 2)(x + 3)} + \frac{(x - 2)(x + 3)}{(x - 2)(x + 3)} > 0.$$

Expanding the numerators and combining all the fractions gives

$$\frac{x^2 + x + 4}{(x - 2)(x + 3)} > 0.$$

Unfortunately, we can't factor the quadratic in the numerator easily. Its discriminant is $1^2 - 4(1)(4) = -15$, so its roots are imaginary. As we saw in Problem 15.3, this means that the quadratic is either always positive or always negative. For a quick review of why, consider the graph of $y = x^2 + x + 4$. Because the roots of $x^2 + x + 4 = 0$ are not real, we know that the graph of $y = x^2 + x + 4$ does not pass through any points with y -coordinate equal to 0. Therefore, the graph does not intersect the x -axis, so it is entirely on one side of the x -axis. Since the graph of $y = x^2 + x + 4$ is an upward-opening parabola that never hits the x -axis, we know that it must be entirely above the x -axis. So, $x^2 + x + 4 > 0$ for all values of x .

Concept: Considering the graph of an expression can help you understand its behavior.



Returning to our problem, we must solve the inequality

$$\frac{x^2 + x + 4}{(x - 2)(x + 3)} > 0.$$

⌘

We know the numerator is always positive. The denominator is positive when $x - 2$ and $x + 3$ are both positive or both negative. They are both positive when $x > 2$ and both negative when $x < -3$, so our solution is $x \in (-\infty, -3) \cup (2, \infty)$. □

We now have a strategy for solving inequalities involving many factors.

4)

Solution: We factor the quadratic to find $(2x - 3)(x + 3)(x + 5) \leq 0$. The product on the left side is negative when all three factors are negative, and when one factor is negative and the other two are positive. We have $2x - 3 < 0$ when $x < 3/2$, $x + 3 < 0$ when $x < -3$, and $x + 5 < 0$ when $x < -5$. We can combine this information in a table:

	$2x - 3$	$x + 3$	$x + 5$	$(2x - 3)(x + 3)(x + 5)$
$x > 3/2$	+	+	+	+
$-3 < x < 3/2$	-	+	+	-
$-5 < x < -3$	-	-	+	+
$x < -5$	-	-	-	-

We see that one factor is negative when $-3 < x < 3/2$ and all three are negative when $x < -5$. Since the inequality is nonstrict, the values $x = 3/2$, $x = -3$, and $x = -5$ are also solutions, so the values of x that satisfy the inequality are $-3 \leq x \leq 3/2$ or $x \leq -5$. In interval notation, this is $x \in (-\infty, -5] \cup [-3, 3/2]$.

5)

Solution: We factor the quadratic as a difference of squares, and we find that we seek all values of r such that $(r - 3)^2(2r - 5)(2r + 5) > 0$. The factor $(r - 3)^2$ is positive for all values of r except $r = 3$. We have $2r - 5 > 0$ for $r > 5/2$, and $2r + 5 > 0$ for $r > -5/2$. So, we see that all factors are positive for all values of r greater than $5/2$ except for $r = 3$. We also have two negative factors when $r < -5/2$, so the product is positive when $r < -5/2$, as well. So, the values of r that satisfy the inequality are

$$\text{all } r > 5/2 \text{ except } r = 3, \text{ and all } r < -5/2.$$

We can write this with intervals as

$$r \in (-\infty, -5/2) \cup (5/2, 3) \cup (3, \infty).$$

6)

Solution: As our first step, we can change the $1 - x$ in the denominator of the first term to $x - 1$. (We don't have to do this, but it will help keep us from making mistakes later.) We have

$$-\frac{2x}{1-x} = \frac{2x}{-1(1-x)} = \frac{2x}{x-1}.$$

So, we can replace the $-\frac{2x}{1-x}$ on the left with $+\frac{2x}{x-1}$, and we have

$$\frac{1}{x-7} + \frac{2x}{x-1} \geq 2.$$

We bring all terms to the left and find a common denominator, which gives us:

$$\begin{aligned} & \frac{1}{x-7} + \frac{2x}{x-1} - 2 \geq 0 \\ \Rightarrow & \frac{x-1}{(x-7)(x-1)} + \frac{2x(x-7)}{(x-7)(x-1)} - \frac{2(x-7)(x-1)}{(x-7)(x-1)} \geq 0 \\ \Rightarrow & \frac{x-1}{(x-7)(x-1)} + \frac{2x^2-14x}{(x-7)(x-1)} - \frac{2x^2-16x+14}{(x-7)(x-1)} \geq 0 \\ \Rightarrow & \frac{3x-15}{(x-7)(x-1)} \geq 0. \end{aligned}$$

We can factor a 3 out of $3x - 15$ to get $3(x - 5)$, then divide both sides by 3 to get

$$\frac{x-5}{(x-7)(x-1)} \geq 0.$$

We can make a table of the signs of our factors:

	$x-7$	$x-5$	$x-1$	$\frac{x-5}{(x-7)(x-1)}$
$x > 7$	+	+	+	+
$5 < x < 7$	-	+	+	-
$1 < x < 5$	-	-	+	+
$x < 1$	-	-	-	-

We see that our expression is positive for $x > 7$ and for $1 < x < 5$. Since the inequality is nonstrict, we must include the values of x that make the expression on the left side of the inequality equal to 0. Only $x = 5$ makes our expression 0; make sure you see why $x = 1$ and $x = 7$ do not. So, our solutions are all x such that $\boxed{x > 7 \text{ or } 1 < x \leq 5}$. In interval notation, this is $x \in (1, 5] \cup (7, \infty)$.

7)

Solution: We bring all the terms to the left, which gives

$$\begin{aligned} \frac{x+3+\frac{1}{x+3}}{(x-3)(x+2)} - \frac{2}{(x-3)(x+2)} &\geq 0 \\ \Rightarrow \frac{x+3+\frac{1}{x+3}-2}{(x-3)(x+2)} &\geq 0. \end{aligned}$$

To simplify this, we find a common denominator of all the terms in the numerator:

$$\begin{aligned} x+3+\frac{1}{x+3}-2 &= x+1+\frac{1}{x+3} \\ &= \frac{(x+1)(x+3)+1}{x+3} \\ &= \frac{x^2+4x+4}{x+3}. \end{aligned}$$

So, our inequality now is

$$\begin{aligned} \frac{\frac{x^2+4x+4}{x+3}}{(x-3)(x+2)} &\geq 0 \\ \Rightarrow \frac{(x+2)^2}{(x+3)(x-3)(x+2)} &\geq 0 \\ \Rightarrow \frac{x+2}{(x+3)(x-3)} &\geq 0. \end{aligned}$$

Even though $x = -2$ makes the left side of this final inequality equal to zero, we cannot include it among our solutions, because it makes a denominator equal to 0 in the original inequality. We turn to a table to determine the values of x that make the left side of our final inequality positive.

	$x-3$	$x+2$	$x+3$	$\frac{x+2}{(x+3)(x-3)}$
$x > 3$	+	+	+	+
$-2 < x < 3$	-	+	+	-
$-3 < x < -2$	-	-	+	+
$x < -3$	-	-	-	-

The expression $\frac{x+2}{(x+3)(x-3)}$ is positive for $x > 3$ (all three factors are positive) and for $-3 < x < -2$ (two negative factors, one positive). Even though the inequality is nonstrict, we can't have x equal to 3, -2, or -3, since each would cause division by 0 in the original inequality. Therefore, the values of x that satisfy the original inequality are all values of x such that $-3 < x < -2$ or $x > 3$. In interval notation, this is $x \in (-3, -2) \cup (3, \infty)$.